ASSIGNMENT FINAL REPORT

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I. Introduction

In today's data-driven economy, data and information are no longer just supporting elements but have become critical assets that shape how organizations operate and compete. The ability to collect, process, and transform raw data into meaningful insights enables businesses to understand market dynamics, predict customer behaviour, and improve internal efficiency. As a result, organizations can make more informed decisions, optimize their business processes, and maintain a strong competitive advantage in an increasingly complex environment.

For global corporations such as Samsung Electronics, data plays a central role across multiple business functions, including supply chain management, demand forecasting, production monitoring, and customer relationship management. The integration of data science tools and technologies further enhances these capabilities by enabling real-time analysis, predictive modelling, and automated decision-making. However, alongside these benefits, the use of data also introduces significant challenges, including social, legal, and ethical concerns, as well as security risks that must be carefully managed.

This report aims to provide a comprehensive analysis of how data and information support real-world business processes. It begins by discussing how data is generated, processed, and utilized within organizations, along with the tools used to transform raw data into actionable insights. The report then evaluates the value of data for both individuals and organizations, and examines the broader social, legal, and ethical implications associated with its use, including common threats and mitigation strategies.

Furthermore, the report explores the role of data science in enhancing decision-making, assessing its benefits in solving practical business problems. A data science solution is then designed and implemented to address a real-world challenge within Samsung's operations, followed by justified recommendations based on analytical findings. Finally, the report evaluates the effectiveness of the applied data science techniques in meeting both user and business requirements.

Through this structured approach, the report highlights not only the importance of leveraging data effectively but also the need for responsible, secure, and strategic data management to support sustainable business success.

II. Contents

P1. Discussion on how data and information support business processes, including the value they have for organizations

1. What is business data?

To answer the question "What is business data?", we first need to understand what data is:

Data is the raw form of information, a collection of facts, figures, symbols or observations that represent details about events, objects or phenomena. By itself, data may appear meaningless, but when organized, processed and interpreted, it transforms into valuable insights that support decision-making, problem-solving and innovation. (GeeksForGeeks, 2026)

In a business context, this has the following significance:

Business data refers to any information related to the operations, performance, customers, or environment of your business. It can come from a variety of sources, – sales figures, customer reviews, website traffic, employee performance, financial reports, inventory levels, and even social media interactions. (Tanny Technologies, 2025)

For example: "100", "Product A", "04/04/2026". Looking at them individually, we wouldn't know if 100 represents the inventory quantity, price, or number of customers.
Characteristics: Objective, discrete, and usually stored in databases.

2. What is information in business?

Information is processed data that is organized, structured, and meaningful, providing context and relevance for decision-making. Unlike raw data, information is actionable and aids in understanding situations, solving problems, or making informed decisions. It is created by analysing and interpreting data to extract patterns, trends, or insights. (Theintactone, 2024)

For example: From the raw data above, we have the information: "On April 4, 2026, we sold 100 units of Product A." This information helps managers understand the sales situation.

3. What types of information in business processes?

In a business, information is typically categorized based on management level and purpose:

- **Strategic Information:** Strategic information is critical for top management as it relates to long-term planning and decision-making. This type of information helps organizations set goals, allocate resources, and navigate complex environments. Strategic information often includes market forecasts, competitor analyses, and insights into economic trends. For instance, a company looking to expand into new markets will rely on strategic information to assess potential opportunities and risks, guiding its long-term vision.

- **Tactical Information:** Tactical information supports mid-level management in implementing the organization's strategies. It is concerned with short- to medium-term decision-making, guiding departments in achieving specific goals aligned with the overall strategic plan. This type of information often includes budget reports, performance metrics, and market analyses. For example, a marketing department may analyse customer feedback and market trends to refine its promotional strategies and improve sales performance.

- **Operational Information:** Operational information pertains to the day-to-day functioning of an organization. This type of information is essential for routine management tasks and includes data related to production, sales, inventory levels, and employee performance. Operational information helps managers monitor ongoing activities and make quick decisions to ensure smooth operations. Examples include daily sales reports, inventory records, and employee attendance logs.

4. How do information and data support business processes?

Data and information act as the lifeblood that keeps processes running smoothly:

- **Decision Making:** Providing factual evidence instead of subjective predictions.
- **Process Optimization:** Helping identify bottlenecks. For example, production time data helps find which step is slowing down the entire line.
- **Automation:** Accurate input data enables software systems (such as ERP, CRM) to automatically execute purchase orders or provide customer service.
- **Communication and Coordination:** Consistent information helps departments (Accounting, Warehouse, Sales) understand each other and work in sync.

5. The values of data and information in the organization

An organization that possesses high-quality data will have the following outstanding advantages:

Type of value	**Detailed description**
Competitive advantage	Gain a better understanding of your customers than your competitors through behavioural data.
Cost savings	Reduce waste by accurately forecasting the amount of inventory needed.
Risk management	Early detection of unusual financial or security issues.
Increase revenue	Personalize the user experience to encourage repeat purchases.
Enhance data-driven decision-making capabilities.	Minimizing guesswork, ensuring strategic choices are based on facts and trends.

Innovation and Growth	Analysing market data and customer feedback can inspire new product ideas or improve existing products.

Table 1. The values of data and information in the organization

P2. Discussion of how data is generated and used by organizations to support business processes and the tools for manipulation to form meaningful data

1. Types of data in organizations and how they are generated

a. Internal Data

Internal data is data generated from within an organization through its day-to-day operations. This includes transaction records, personnel information, financial reports, and production data. It is typically stored in databases or other information systems.

- **How it's generated:** It originates from management systems such as ERP (Enterprise Resource Planning) and CRM (Customer Relationship Management).
- **Examples at Samsung:** Sales history of the Galaxy S24 series, inventory levels of components at factories in Bac Ninh, or data on worker performance on the production line.

b. External Data

External information refers to data collected from sources outside the organization's purview. This type of data helps businesses understand market conditions, customer behaviour trends, and external factors that may impact business operations. Typical examples include market demand forecasts, supplier data, economic indicators, and industry analysis reports.

- **How to generate it:** Through market surveys, data collection from websites (web scraping), or purchasing data from data research companies.
- **Examples at Samsung:** Competitor semiconductor component prices, user reviews on YouTube, or changes in global consumer electronics purchasing habits.

c. Generate data in organization

Data doesn't just appear out of nowhere; it's generated through various activities and interactions both inside and outside an organization. Every action, transaction, or process can become a valuable data source:

- **Customer interactions:** Every time a Samsung TV's electronic warranty is activated, a new data point is recorded. Additionally, activities like purchases, service feedback, product reviews, or contacting customer support contribute to a large amount of behavioural and demand data.

- **Machine operations (IoT):** In smart factories, IoT sensors and devices continuously collect and transmit data on machine status, performance, temperature, vibration, etc. This data helps businesses monitor, predict errors, and optimize production processes.

- **Business processes:** Activities such as signing contracts with component suppliers, processing orders, managing inventory, and making payments all generate data related to finance, legal matters, and operations. This is a crucial data source for internal control and decision-making.

- **Internal operations:** Daily tasks such as employee timekeeping, performance evaluations, internal email communication, or the use of management software continuously generate data for human resource and organizational management.

- **Marketing and communication activities:** Advertising campaigns, social media interactions, website traffic, and user behaviour on digital platforms all generate data that helps businesses analyse the effectiveness of their marketing strategies and make necessary adjustments.

In general, data within an organization is generated continuously and in diverse ways from many different sources. Understanding how data is generated will help businesses manage, exploit, and leverage data more effectively in decision-making and long-term development.

2. Tools for Data processing

To transform raw data into valuable information for the supply chain, appropriate processing tools are needed. These tools help collect, clean, analyse, and visualize data more efficiently.

a. Some tools for data processing

Currently, many tools, from basic to advanced, are used in data processing:

- **Microsoft Excel:** The most popular tool, suitable for quick calculations, managing small-scale data, and creating simple tables.
- **Tableau:** Known for its powerful data visualization capabilities, transforming complex data into easy-to-understand charts.
- **SQL (Structured Query Language):** Used to query, manipulate, and manage data in database management systems.
- **Google Analytics:** Supports tracking and analysing user behaviour on websites, very useful in the digital business environment.

In addition, other tools such as **Power BI** or **ERP** systems are often integrated to expand data analysis capabilities within businesses.

b. Programming language for data processing

When available tools are insufficient, analysts will use programming languages to process data more flexibly:

- **R:** Very powerful in statistical analysis, data modelling, and academic research.
- **Python:** Considered the mainstay language in the data field thanks to its rich library ecosystem, including Pandas and NumPy, supporting everything from data processing to machine learning.

Combining programming with visualization tools significantly increases the efficiency of data processing and analysis.

c. What tools are used in my project?

In my supply chain optimization project for Samsung, I chose to combine two main tools:

- **Python:** Used as a backend processing tool. Python helps with data cleaning, removing duplicate data, handling errors, and standardizing data from various sources before analysis. Additionally, Python supports automation of several processing steps, saving time.

- **Microsoft Power BI:** The primary tool for building dashboards. Power BI connects to multiple data sources, allowing for the creation of intuitive charts and dynamic reports. This enables management to easily monitor inventory levels, analyse trends, and forecast demand in real time.

The combination of Python and Power BI creates a complete data processing workflow, from data preparation to presenting results in a visual and easy-to-understand manner.

d. Power BI

Within the scope of this project, Microsoft Power BI was applied as a central tool to target and optimize initial data into valuable service-oriented analytics. Previously, data from various sources such as sales transactions, product catalogues, customer information, inventory, and market trends were integrated into the system. Then, links between tables were established to form a complete model, ensuring optimal functionality and seamless analytical capabilities across data components.

Once the configured data was properly organized, a system dashboard was built to display key metrics. These metrics included not only overall revenue but also extended to product consumption volume, customer conversion levels, and market trends by region. Simultaneously, comparative charts between actual data and quarterly (quarterly) data reports were designed to support the assessment of forecast accuracy and optimization goals.

Through the application of Power BI, data is no longer fragmented but organized into a visual, easy-to-understand system. This helps businesses continuously monitor operations, quickly detect problems, and make evidence-based decisions, thereby improving supply chain management efficiency.

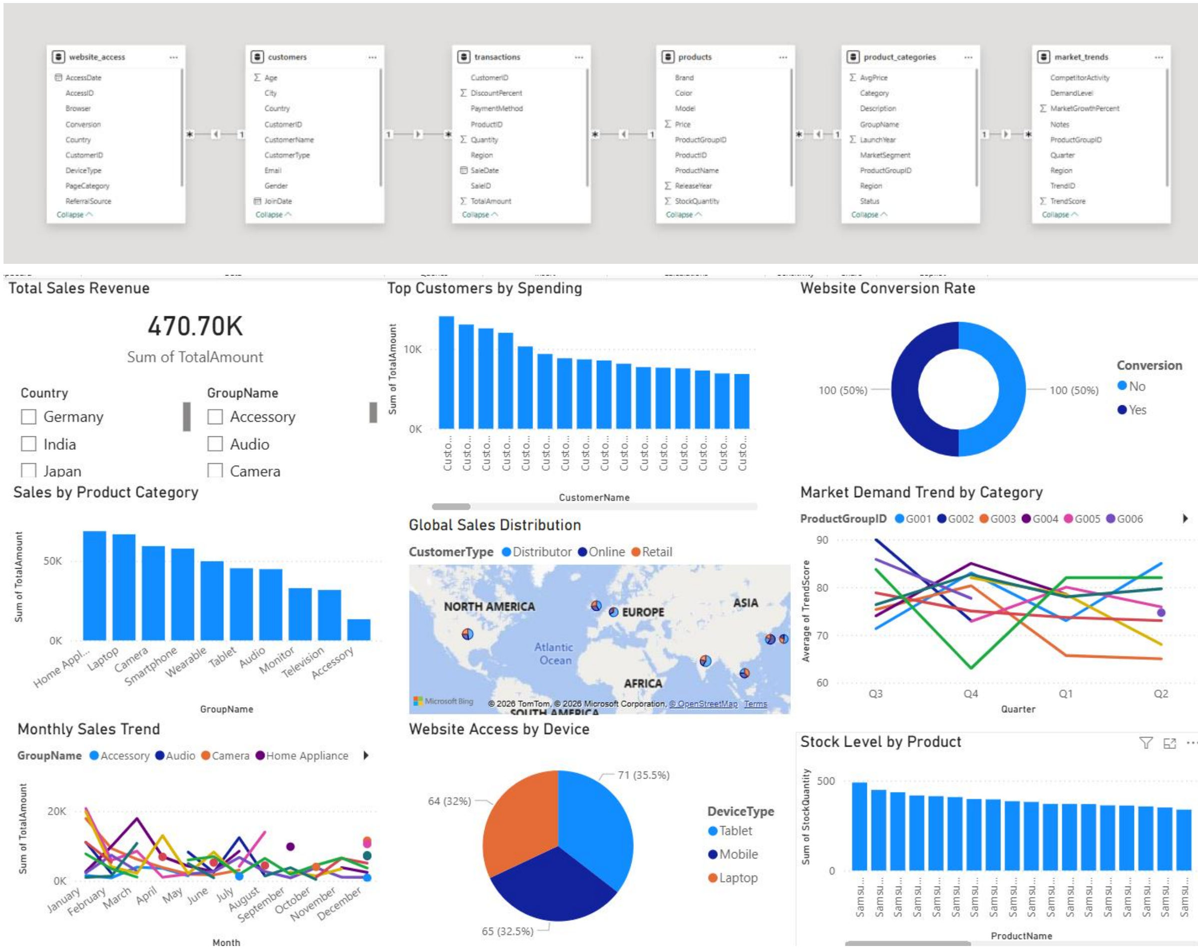


Figure 1. Power BI Dashboard

M1. Assess the value of data and information to individuals and organisations in relation to real-world business processes

To gain a comprehensive and objective perspective, data and information should be viewed as strategic assets, not merely isolated numbers. In the modern business environment, data plays a crucial role in minimizing uncertainty, supporting trend prediction, and improving decision-making quality at all levels.

1. Individual

In any organization, individuals (employees, analysts, managers) are the ones who directly exploit and use data in their daily work. Proper data access offers many practical benefits:

Overall Value:

- **Improved work performance:** Accurate data helps individuals process tasks faster, minimize errors, and reduce the need for repeated revisions. As a result, productivity is significantly increased.

- **Increased confidence in decision-making:** Instead of relying on intuition, employees can use data to make well-founded proposals, increasing persuasiveness when working with superiors or partners.

- **Developed personal capabilities:** Regularly interacting with data helps individuals hone their analytical thinking, logical reasoning abilities, and problem-solving skills in a real-world environment.

- **Enhanced Adaptability:** In the context of digital transformation, individuals with data skills will easily adapt to new technologies and increasingly complex job requirements.

At Samsung Electronics:
As an intern, data helps I quickly understand the relationship between customer behavior and the national data warehouse system. Instead of time-consuming manual processing, scientifically organized data allows you to identify problems, propose optimal solutions, and contribute more effectively to projects. This not only improves personal performance but also creates opportunities for recognition from management, such as the Head of Data Analyst, opening up the possibility of participating in more important projects in the future.

2. Organisation

At the organizational level, especially for multinational corporations, data is considered the backbone of sustainable operation and development. The value of data is clearly demonstrated in the following aspects:
Overall Value:

- **Creating a Competitive Advantage:** Businesses that leverage customer data more effectively will be able to grasp market needs faster, thereby gaining an advantage over competitors.

- **Optimizing Operating Costs:** Data helps identify inefficiencies in processes, thereby optimizing resources and reducing unnecessary costs.

- **Effective Risk Management:** Through the analysis of historical data and trends, organizations can predict fluctuations and prepare timely response plans.

- **Supporting Long-Term Strategy:** Data not only serves short-term operations but also helps build sustainable development strategies based on concrete evidence.

At Samsung Electronics:
In the context of complex global supply chain management, data plays an extremely important role:

- **Demand Forecasting:** By analyzing sales data and market trends, Samsung can flexibly adjust production, avoiding product shortages or surpluses.

- **Inventory Optimization:** Data helps balance avoiding stockouts and limiting excess inventory, thereby saving storage and operating costs.

- **Enhancing Customer Experience:** Ensuring products are always available when customers need them not only increases revenue but also strengthens brand reputation.

- **Increasing Operational Transparency:** Departments can access the same data source, helping to synchronize information and reduce conflicts in the decision-making process.

3. In my project

The project of building a dashboard using Microsoft Power BI is not only academically valuable but also has high practical application value in a business context.

Potential values of the analytical tool:

- The ability to connect and synchronize data from multiple sources in near real-time.
- Transforming complex data into intuitive, easy-to-understand charts.
- Supporting remote access, allowing multiple departments to use it simultaneously.

- Increased data interaction through dynamic dashboards.

Assessment of practical value in the project:
In this project, I focused on linking customer data with datastore systems. The core value that the solution brings is transparency and timeliness of information:

- The dashboard helps transform raw data into immediately understandable insights, helping users quickly grasp the situation.
- Instead of waiting for periodic reports, users can directly monitor changes in the supply chain in real time.
- Anomalies (e.g., declining sales or sudden spikes in inventory) can be detected early and addressed promptly.

Therefore, the solution acts as an effective decision-making support tool, helping Samsung Electronics implement its Best Practice strategy, ensuring all production and distribution coordination activities are based on real data rather than guesswork.

P3. Discuss the social legal and ethical implications of using data and information to support business processes

The exploitation and use of data goes beyond simply improving business efficiency; it is directly related to human rights, social norms, and legal compliance. As data becomes an increasingly important asset, the accompanying implications are also becoming more complex and require a more comprehensive consideration.

1. The social implications of using data and information in business

Figure 2. Social

Data is reshaping how people interact, work, and make decisions in modern society.

- **Personal privacy:** The large-scale collection and analysis of data can lead to individuals feeling constantly monitored, creating what is known as a "surveillance society." This impacts personal freedom and online behavior.

- **The digital divide:** Individuals lacking access to technology or digital skills will have limited opportunities in education, employment, and business, thus increasing social inequality.

- **Changes in the labor structure:** The application of data and automation increases production efficiency but simultaneously reduces the need for unskilled labor. Conversely, the demand for positions such as Data Analyst and Data Scientist is increasing sharply, creating a shift in the occupational structure.

- **Manipulating user behavior:** Digital platforms use algorithms to personalize content, easily leading to "filter bubbles," causing users to only access information that aligns with their preconceived views, reducing diversity and potentially causing social division.

- **Impact on public perception:** Data and algorithms can prioritize the dissemination of attention-grabbing content, sometimes misinformation, thereby negatively impacting public perception and behavior.

2. The legal implications of using data and information in business

Figure 3. Legel

This is a mandatory aspect, requiring organizations to strictly comply to avoid legal risks.

- **Personal data protection laws:** Regulations such as the General Data Protection Regulation (GDPR) in Europe or Decree 13/2023/ND-CP in Vietnam require businesses to collect, store, and process data transparently and securely.

- **Data ownership:** Determining who is the true owner of data (users or businesses) remains a contentious issue, especially in data-driven business models.

- **Data breach liability:** When data is leaked or attacked, businesses not only suffer reputational damage but may also face significant fines and compensation obligations to customers.

- **International compliance:** For multinational corporations, data processing must also comply with various legal systems, increasing the complexity of governance and operations.

3. The ethical implications of using data and information in business

Figure 4. Ethical

Besides legal factors, ethical considerations play a guiding role in what businesses should do, even when specific regulations are not yet in place.

- **Algorithmic Bias:** If input data is biased, the analysis system or AI may produce unfair results, negatively impacting certain groups of people.

- **Transparency:** Businesses need to publicly disclose how they collect and use data. Hiding information in long and confusing terms and conditions can erode user trust.

- **Consent:** Users need to be given clear and voluntary choices when sharing data, rather than being forced through default or misleading mechanisms.

- **Ethical data exploitation:** Even when legal, using data for purposes such as manipulating consumer behavior or discrimination (e.g., denying services based on health data) raises many questions about its appropriateness.

- **Corporate social responsibility:** Businesses need to consider the long-term impact of data use on the community, rather than focusing solely on short-term profits.

4. Why must we have regulator for using data and information in business?

In the context of an increasingly complex data war, regulatory bodies act as intermediaries to ensure fairness, transparency, and security for the entire digital ecosystem. Organizations such as the Ministry of Information and Communications and consumer protection associations not only monitor but also guide how data is used in modern society.

- **Balancing power:** Large technology corporations often hold vast amounts of data, creating a significant advantage over individuals or small businesses. This can lead to an imbalance of power. Regulatory bodies help protect vulnerable users by limiting data abuse and ensuring privacy is respected.

- **Establishing common standards:** Establishing common regulations and standards helps all businesses operate on a level playing field. This not only creates a healthy competitive environment but also ensures that data is processed according to consistent principles, minimizing risks and errors in the operational process.

- **Enforcing discipline:** Without regulatory bodies, legal regulations would be difficult to enforce effectively. These organizations have the authority to inspect, evaluate, and apply penalties for violations, thereby ensuring that businesses strictly comply with regulations on data and information security.

- **Protecting national security: Big** data not only has economic value but is also directly related to digital sovereignty and national security. Controlling the flow of data, especially cross-border data, helps prevent risks such as information leaks, cyberattacks, or illegal data exploitation.

- **Enhancing Social Trust:** In addition to the roles mentioned above, regulatory bodies also contribute to building user trust in the digital environment. When people know that their data is protected by clear regulations and strict oversight mechanisms, they will be more willing to participate in and contribute to the digital economy.

Overall, the presence of regulatory bodies is indispensable to ensuring that data use occurs responsibly, fairly, and sustainably in the digital age.

P4. Describe common threats to data and how they can be mitigated on a personal and organisational level

1. Identify common threats to data and information

Figure 5. Common threads

A cyberattack is an attempt to steal, alter, destroy, disrupt, or disable information resources and systems found in computer networks and systems. Cyberattacks can fit into two categories: insider threats or outsider threats. Insider threats stem from individuals with legitimate access to the systems they target, using their access to exploit vulnerabilities intentionally or inadvertently. They could be committed by a dissatisfied or angry employee or a contractor with access to the organization’s systems. An outsider threat is from someone who doesn’t have any affiliation with the system they’re attacking, such as criminal organizations or hackers. (Coursera, 2025)

Data can be compromised due to a variety of reasons, including technical errors, human error, or targeted external attacks. In the digital age, threats are becoming increasingly sophisticated and difficult to detect, requiring both individuals and organizations to have a clear understanding to prevent them.

a. Personal level

At this level, attackers typically target digital assets, personal accounts, and user identities:

- **Phishing:** Emails or messages impersonating banks, social media, or online services to trick users into clicking on malicious links or providing login information such as passwords or OTP codes.

- **Malware:** Includes viruses, trojans, spyware, or adware, often spread through cracked software, files downloaded from unknown sources, or insecure websites.

- **Weak or shared passwords:** Using simple passwords (e.g., 123456, date of birth) or using the same password for multiple accounts significantly increases the risk of information theft.
- **Social Engineering:** Attackers exploit trust, impersonating acquaintances or reputable organizations to trick users into providing sensitive information.

- **Personal data leaks:** Sharing too much information on social media can also lead to it being exploited for targeted attacks.

b. Organisation level

At the enterprise level, threats are larger in scale and more damaging:

- **Ransomware:** Attackers encrypt entire data systems and demand a ransom for restoration, potentially disrupting entire business operations.

- **Insider Threats:** Employees may inadvertently leak data due to lack of knowledge, or intentionally steal information for personal gain.

- **DDoS (Denial-of-Service) Attacks:** Systems are overloaded by fraudulent traffic, preventing services from functioning properly.

- **SQL Injection:** Attacks exploiting security vulnerabilities in websites to access, modify, or steal data from databases.

- **Lack of Security Configuration:** Systems that are not properly configured (misconfiguration) are also a common cause of data leaks.

2. How the threats can be mitigated

Figure 6. Cyber Security

While it's impossible to completely eliminate all risks, we can significantly mitigate them by implementing appropriate security measures.

a. Personal level

- **Use multi-factor authentication (MFA/2FA):** Enable two-factor authentication for important accounts such as email, social media, and banking to enhance protection.

- **Manage passwords effectively:** Use a password manager to create and store complex passwords, avoiding manual memorization or sharing passwords.

- **Update software regularly:** Always keep your operating system and applications on the latest version to patch security vulnerabilities.

- **Back up data:** Store important data on the cloud or external devices to ensure recovery in case of problems.

- **Be cautious when accessing:** Avoid clicking on suspicious links, do not download software from unknown sources, and carefully check information before providing personal data.

b. Organisation level

- **Establish a multi-layered security system:** Utilize firewalls and intrusion detection and prevention systems (IDS/IPS) for comprehensive system protection.

- **Data encryption:** Apply encryption to data during storage and transmission to prevent unauthorized access.

- **Least Privilege Principle:** Grant only necessary access to each employee, minimizing the risk of abuse.

- **Security awareness training:** Conduct regular training programs to help employees recognize threats such as phishing and enhance security awareness.

- **Disaster Recovery Plan:** Develop a rapid response process to ensure the system can recover quickly in the event of an incident.

- **Regular monitoring and inspection:** Conduct regular security checks to detect vulnerabilities early and address them promptly.

M2. Analyse the impact of using data and information to support business real-world business processes

1. Project Background

During my three-month internship at Samsung Electronics, I participated in projects related to connecting customers (including retail partners and distributors) with datastore systems at the local and national levels. This was a highly practical work environment where data played a central role in all analytical and decision-making activities.

The biggest challenge Samsung faced lay in the complexity of its global supply chain. With a diverse product portfolio ranging from smartphones and tablets to home appliances, imbalances between supply and demand could lead to two serious consequences: excess inventory resulting in wasted storage costs or shortages causing lost business opportunities. Therefore, the main objective of the project was to build a synchronized information system that would enable the company to make quick and accurate decisions based on real-world data rather than intuition.

2. How data and Information were used

In this project, data is not only collected and stored but also processed and exploited as a forecasting tool and strategic support:

Data Warehouse Connectivity: The system is designed to retrieve data from various sources such as national data warehouses on taxes, transportation, and consumer behavior by region. This helps create a comprehensive picture of the market.

Historical Data Analysis: Past sales figures are used to identify seasonal trends and consumption cycles. For example, demand for TVs often surges before major sporting events, helping businesses prepare in advance for production and distribution.

Real-time Data: Through API integration with dealers, the system can continuously update inventory status. This allows for the immediate detection of products in high demand or at risk of running out of stock.

Combining multiple data sources: Integrating internal and external data helps improve the accuracy of analysis and forecasting.

3. Impact on Business Processes

The shift from an experience-based to a data-driven management model has brought about significant changes throughout the entire operational process:

- **Accurate Demand Forecasting:** Instead of mass production based on estimates, Samsung can adjust production based on specific forecasts. This significantly improves inventory turnover and minimizes inventory risk.

- **Resource Optimization:** Data helps the business allocate manpower, raw materials, and production capacity to high-demand products, thereby improving resource efficiency.

- **Improved Supply Chain:** Thanks to early forecasting information, the shipment of components from suppliers to the factory is planned more accurately, minimizing delays and additional costs.

- **Increased Responsiveness:** When the market fluctuates, the data system allows the business to adjust its strategy almost immediately instead of spending days on manual analysis.

When data helps reduce inventory costs, optimize operations, and simultaneously increase sales, net profit increases while investment costs are controlled. This leads to a significant improvement in the ROI of the entire business process.

4. Real-World Example

A prime example is the Samsung Galaxy S-series product launch process:

- **Before applying data:** Businesses might produce large quantities of color or configuration versions that don't match market demand, leading to excess inventory and the need for price reductions to clear stock.
- **After applying data:** By analyzing customer preferences from pre-order and retail data, Samsung identifies prominent trends such as popular colors or versions.

- **Result:** The company adjusts its production plan to match actual demand, ensuring that hot versions are always in stock (reducing stockouts) and limiting the overproduction of less popular versions (reducing excess inventory). This not only optimizes revenue but also enhances the customer experience.

5. Lessons Learned and Ongoing Impact

From this project, I've drawn several important lessons that can be applied to future organizations and projects:

- **Data is a dynamic entity:** Data needs to be constantly updated. In the technology industry, information quickly becomes outdated if it's not regularly refreshed.

- **Connectivity is key:** Successful integration with national datastores gives businesses a holistic view of the market, far beyond what internal data can provide.

- **The importance of visualization:** Data is only truly valuable when presented in an easily understandable way. Using Microsoft Power BI to build dashboards helps managers quickly grasp information and make decisions in a short time.

- **Long-term impact:** Building a data-driven decision-making culture not only provides short-term benefits but also helps businesses maintain a sustainable competitive advantage in the future.

Overall, the use of data and information has brought about a comprehensive change in how businesses operate, from the strategic level to day-to-day operations, helping to improve efficiency, reduce risk, and increase adaptability to the market.

D1. Evaluate the wider implications of using data and information to support business processes in an identified organisation

1. Using tools and techniques to process data within the organization

To transform data into a valuable asset, Samsung Electronics goes beyond simply collecting it and invests heavily in advanced processing tools and techniques. This process includes data cleaning, multi-source integration, and visualization to facilitate rapid decision-making.

The Importance of Visualization:
Humans tend to process visual information much faster than text. Therefore, a well-designed dashboard can help managers identify problems in seconds, instead of having to read thousands of lines of raw data. This is especially important in a fast-changing business environment.

A practical example using Microsoft Power BI at Samsung:

- **Techniques applied:** Data from the Sales, Inventory, and Distribution departments are integrated into a single dashboard system. Metrics such as sales by region, inventory levels, and sell-through rates are updated in real time.

- **Advanced Analysis:** The system uses heatmaps to display areas with supply-demand imbalances. For example, some provinces have high Galaxy S24 inventory but low sales, while other areas experience shortages.

- **Practical Action:** Upon identifying the problem, the Marketing department can implement regional promotions, adjust prices, or redistribute goods among dealers. Simultaneously, the Logistics department can shift inventory to areas with higher demand.

- **Concrete Results:** Response time is shortened from weeks to hours, significantly reducing inventory costs and increasing short-term revenue. This clearly demonstrates the value of real-time data visualization.

2. Combining data generated by humans and machines

One of Samsung's core strengths is its ability to effectively combine two types of data:

- **Machine-Generated Data:** This includes data from IoT sensors in chip manufacturing plants, system logs from websites, and usage data from smart devices. This is a highly accurate, large-volume, and continuously updated data source.

- **Human-Generated Data:** This includes customer feedback, product reviews, comments from sales staff, and strategic decisions from management. This data is contextual, emotional, and reflects the actual behavior of users.

By combining these two data sources, Samsung not only knows what is happening but also understands why it is happening. For example, the system might record a decline in sales in a particular area (machine data), but through customer feedback and social media (human data), the company discovers that the cause is a new trend on a platform like TikTok prioritizing a competitor's product.

Significance: This collaboration allows Samsung to respond more quickly and accurately, such as adjusting marketing strategies or improving products.

3. Positive Implications

The application of data creates many positive systemic impacts:

a. Improved Decision-Making

Data-driven decisions reduce reliance on personal feelings and increase objectivity.

Accuracy Rate = (Total Decisions/Successful Decisions) × 100

Data provides clear evidence, helping to improve the accuracy rate of decisions and minimize the risk of errors.

b. Enhanced Operational Efficiency

Through supply chain data analysis, Samsung applies the Just-in-Time (JIT) model, reducing unnecessary inventory and optimizing logistics processes. This helps reduce warehousing costs and increase the speed of goods turnover.

c. Targeted Marketing and Customer Engagement

Data allows for more detailed customer segmentation. For example, users searching for energy-saving products will be shown relevant ads, increasing conversion rates and marketing campaign effectiveness.

d. Risk Assessment and Mitigation

Through data, Samsung can predict risks such as supply chain disruptions or component shortages, thereby preparing timely alternative solutions.

e. Innovation and Creation of New Opportunities

User data helps identify new trends, such as the demand for foldable screen devices, thereby driving the development of innovative product lines.

f. Privacy and Data Security

Investing in security systems like Samsung Knox increases customer trust and creates a sustainable competitive advantage.

g. Organizational culture and change management

The use of data fosters a data-driven culture, where all employees rely on data to work, making organizations more flexible and adaptable to the market.

4. Negative Implications

Besides the benefits, over-reliance on data also brings significant risks:

- **High investment costs:** Building a data system, purchasing tools like Power BI, and hiring data experts require substantial financial resources.

- **Security risks:** The more data there is, the higher the risk of attack. A data leak can severely damage a company's reputation.

- **Analysis paralysis:** Too much data can cause managers to hesitate in making decisions, slowing down their response to the market.

- **Technology dependence:** If the system fails, the entire decision-making process can be disrupted.

- **Ethical issues:** The collection and analysis of user behavior data, if not transparent, can be offensive and erode customer trust.

P5. Discuss how tools and technologies associated with data science are used to support business processes and inform decisions

1. Introduction to Data Science and Its Role in Business

Data science is the study of data used to extract meaningful insights for business decisions. It combines mathematics, computing and domain knowledge to solve real-world problems and uncover hidden patterns. (GeeksForGeeks, 2026)

In the modern business environment, especially at Samsung Electronics, data not only supports operations but also shapes long-term strategy. Instead of relying on personal experience, managers use data models to:

- **Minimize risks:** Predict market fluctuations and prepare contingency plans.

- **Increase operational efficiency:** Optimize production, logistics, and distribution processes.
- **Personalize customer experiences:** Analyze behavior to provide products and services tailored to each customer group.
- **Support rapid decision-making:** Provide real-time information to help leaders respond promptly to change.

2. Popular Tools in Data Science

To process large volumes of data from millions of devices and transactions, analysts often use a combination of tools:

Programming Languages:

- **Python:** Considered the backbone of Data Science thanks to its powerful library ecosystem such as Pandas (data processing), NumPy (computation), Scikit-learn (machine learning), and Matplotlib/Seaborn (visualization). In addition, Python supports automation of the analysis process.

- **SQL:** An essential tool for querying and managing data in large database management systems. SQL helps extract data quickly and accurately from datastores.

Visualization Tools:

- **Microsoft Power BI:** Widely used to build dynamic dashboards, supporting real-time KPI tracking.
- **Tableau:** Suitable for large and complex datasets, allowing the creation of in-depth interactive charts.

Data processing and storage platforms:

- **Apache Hadoop:** Supports large-scale distributed data storage and processing.
- **Apache Spark:** Enables faster data processing thanks to its in-memory parallel computing capabilities.

These tools combine to form a complete ecosystem for data collection, processing, and analysis.

3. Supporting Technologies in Data Science

Besides the tools themselves, technology platforms play a crucial role in expanding data processing capabilities:

Technology	Main role	Main role
Artificial Intelligence (AI) and Machine Learning	Automated learning from historical data to make predictions.	Increase forecasting accuracy and detect risks early.
Internet of Things (IoT)	Collect real-world data from sensors and devices.	Real-time asset and operations monitoring
Cloud Computing	Providing flexible data storage and computing infrastructure.	Cost-effective, easy to scale up.
API & system integration	Connecting and transferring data between platforms.	Ensure data consistency and compatibility throughout the entire system.

Table 2. Supporting Technologies in Data Science

Real-world examples from large corporations:

- **Netflix (AI & Machine Learning):** Uses machine learning algorithms to analyze the viewing behavior of millions of users. From this, their personalized content recommendation system helps retain customers and optimize the cost of producing new films.

- **Rolls-Royce (Internet of Things - IoT):** They install thousands of sensors on their aircraft engines. Data on temperature, pressure, and speed are sent in real time for predictive maintenance, helping the company know exactly when repairs are needed before problems occur.

- **Coca-Cola (Cloud Computing):** This corporation uses cloud platforms (such as AWS and Azure) to consolidate data from thousands of factories and distribution partners globally. This allows them to analyze their supply chain extremely quickly without maintaining huge physical data centers in each country.

- **Uber (API & System Integration):** Uber uses the Google Maps API for location and route calculation, and the Braintree/Stripe API for payment processing. Integrating these systems ensures a seamless data flow from booking to payment completion within a single application.

4. Applications Supporting Business Processes and Decision-Making

At Samsung Electronics, these tools and technologies are directly applied to many operations:

- **Demand Forecasting:** Using Machine Learning models to analyze sales data, market trends, and consumer behavior, thereby providing more accurate forecasts of product demand.
- **Predictive Maintenance:** Analyzing data from IoT sensors to detect anomalies in machinery, helping to minimize downtime and repair costs.
- **Quality Management:** Data collected at inspection points on the production line helps detect defects early, reducing defective product rates and recall costs.
- **Logistics Optimization:** Real-time data helps coordinate transportation more efficiently, reducing costs and delivery times.
- **Strategic Decision Support:** Dashboards aggregate data from multiple sources, giving leaders a comprehensive view and enabling faster decision-making.

5. Real-World Case Studies

Walmart deploys a smart sensor system and data encryption across its entire fresh food network:

- **Detailed process:** IoT sensors attached to packages continuously record temperature, humidity, and GPS location throughout the transportation process from farm to warehouse. This data is directly uploaded to a cloud platform based on Blockchain technology.
- **Data analysis:** The system automatically checks whether environmental indicators are within safe limits. If the temperature rises (risk of food spoilage), an alert is sent immediately.

- **Decision-making:** In case of quality issues or bottlenecks, the system automatically redirects the shipment to the nearest store for urgent consumption or removes the defective shipment before it enters the warehouse.
- **Results:** Reduced traceability time from 7 days to 2.2 seconds, minimizing food waste and ensuring absolute safety for consumers.

Walmart integrates a massive data system with thousands of suppliers (such as P&G, Unilever, Coca-Cola):

- **Visibility:** Through the Retail Link platform, suppliers can directly track sales and inventory levels of their products at each Walmart store in real time.
- **Decision automation:** Walmart uses a Vendor Managed Inventory (VMI) model. When the scanner at the checkout counter records a sale, the Cloud system automatically sends a purchase order to the supplier so they can proactively prepare the goods without waiting for traditional purchase orders.
- **Advanced analytics:** Historical sales data combined with weather forecasts and social events helps the system suggest optimal inventory levels for each season in different geographic areas.
- **Results:** Optimized inventory turnover, significantly reduced warehousing costs, and ensured shelves are always fully stocked (achieving an availability rate of over 95%).

M3. Assess the benefits of using data science to solve problems in real-world scenarios

1. Key Benefits in Real-World Scenarios

In the modern business landscape, data science is no longer a supplementary element but has become a core foundation for businesses to operate efficiently and competitively. Applying data science also brings many clear benefits, which can be listed as follows:

- **Predictability:** Businesses shift from a reactive to a proactive approach. For example, instead of waiting for a sudden surge in demand before increasing production, the system can predict and prepare resources in advance.

- **Accuracy and Objectivity:** Decisions are no longer based on personal experience but on real-world data, reducing errors and bias.

- **Scalability:** Data models can handle massive amounts of data from multiple markets simultaneously, something humans cannot do manually.

- **Resource Optimization:** Data helps accurately identify the most profitable areas, products, or timeframes, thereby allocating resources efficiently.

- **Increased Decision-Making Speed:** Thanks to dashboards and real-time data, leaders can make decisions in minutes instead of waiting for manual reports.

2. Comparison: With and Without Data Science

Characteristic	Without Data Science	With Data Science
Demand forecast	Based on past sales figures and subjective feelings.	Using Machine Learning to analyze trends, behaviors, and external data.
Machinery maintenance	Only fix when it's broken.	Predictive maintenance based on IoT sensor data.
Inventory management	Maintain excess inventory to avoid shortages.	Real-time inventory optimization
Customer feedback	Only process complaints when they arise.	Early error detection from production data and user behavior.
Decision making	Slow, dependent on experience.	Fast, based on real-time data.

Table 3. Comparison: With and Without Data Science

This comparison shows that data science not only improves performance but also completely changes the way businesses operate.

3. Assess the benefits of using Data Science in the scenario

To fully understand the role of Data Science, it's necessary to evaluate it from two perspectives: the value it brings to Samsung Electronics and the risks and limitations if the company doesn't adopt this approach in practice.

a. Benefits of applying Data Science at Samsung

- **Accurate Market Demand Forecasting:** Data science helps Samsung analyze sales data, consumer trends, and seasonality to make accurate forecasts. This helps the company balance supply and demand, avoiding shortages or excess inventory.

- **Optimized Supply Chain:** Real-time data from logistics and warehousing systems helps Samsung coordinate production and transportation more efficiently. As a result, delivery times are shortened and operating costs are significantly reduced.

- **Improved Product Quality:** Through analyzing data from the production line and customer feedback, Samsung can detect errors early and continuously improve products, enhancing the user experience.

- **Support for Fast and Accurate Decision-Making:** Dashboards and analytical tools give leaders a comprehensive overview and enable them to make data-driven decisions rather than relying on intuition.

- **Increased competitiveness:** Understanding customer behavior helps Samsung implement accurate marketing strategies, thereby increasing revenue and maintaining its market position.

b. Consequences of not using Data Science

Inaccurate forecasting: Without data, Samsung would have to rely on experience or intuition, leading to deviations in production and distribution. This could result in large inventories or lost sales opportunities.

- **Inefficient supply chain:** Lack of real-time data causes delays in goods coordination, easily leading to bottlenecks or shortages in critical areas.

- **Increased operating costs:** Failure to optimize resources leads to waste in production, warehousing, and transportation.

- **Slow response to the market:** The business cannot quickly detect new trends or changes in customer behavior, thereby losing its competitive advantage.

- **Quality and reputation risks:** Failure to detect product defects early can lead to major incidents, directly impacting the brand.

c. Overall Assessment

Data science not only brings operational efficiency benefits but also helps Samsung build an intelligent and flexible decision-making system. Conversely, without its application, the company would face numerous risks ranging from costs and performance to competitiveness.

From an overall perspective, it can be affirmed that data science plays a crucial role in helping Samsung Electronics maintain its leading position and achieve sustainable growth in the global market.

P6. Design a data science solution to support decision making related to a real-world problem

1. Identify the problem

In the context of global operations, Samsung Electronics is facing numerous challenges related to supply chain and production management.

- **The core issue:** Inaccurate demand forecasting leads to two common problems: stockouts, resulting in lost sales opportunities, and excess inventory, leading to wasted storage costs.
- **Additional problem:** Sudden machine breakdowns at the factory disrupt production, affecting delivery schedules and increasing maintenance costs.
- **The underlying cause:** Existing data is not fully utilized and remains fragmented across departments (Sales, Logistics, Production), resulting in a lack of a holistic view.
- **Impact:** These issues increase operating costs, reduce resource efficiency, and directly impact customer experience and brand reputation.

2. Goals of the solution

Data Science solutions are designed to transform operational models from passive to proactive and smarter:

- **Predicting future demand:** Building accurate forecasting systems by region, product, and time to optimize production.
- **Predictive maintenance:** Early detection of failures from sensor data to reduce downtime.
- **Resource optimization:** Efficient allocation of materials, manpower, and production capacity based on data.
- **Improved decision-making speed:** Providing real-time information to help leaders react quickly to market changes.
- **Enhancing customer experience:** Ensuring products are always available when customers need them.

3. Design a Data Science Solution

a. Data Sources

The solution requires the integration of diverse data sources to ensure comprehensiveness:

- **Sales and customer data:** Including transaction history, purchasing behavior, demographic information, and product catalog.
- **IoT and sensor data:** Collected from the production line regarding temperature, pressure, machine performance, and energy consumption.
- **Logistics data:** Information on shipping routes, delivery times, and real-time inventory status.
- **External market data:** Consumer trends, social media data, and macroeconomic factors influencing demand.

Data Formats:

- CSV exports from CRM and ads platforms
- SQL databases for transaction logs
- JSON API responses from analytics tools

b. Data Processing Plan

Raw data will be processed through a standardized pipeline to ensure accuracy and consistency:

- **Data Cleaning:** Addressing missing values using mean/median, removing duplicate data, and detecting outliers.

- **Normalization/Encoding:** Converting categorical data into numerical form and bringing variables to the same scale for effective model performance.
- **Feature Engineering:** Creating new variables such as seasonality (month, quarter), holidays, and marketing campaigns to improve forecast accuracy.
- **Data Integration:** Connecting data from multiple sources into a unified system.
- **Data Quality Checking:** Ensuring the reliability of input data before inputting it into the model.

c. Analytical Approach

The solution focuses on exploiting hidden patterns in data:

- **Trend and Seasonal Analysis:** Identifying periods of high demand such as festivals or special events.
- **Pattern Analysis:** Monitoring IoT data to detect anomalies in machine operation.
- **Correlation Analysis:** Evaluating the relationship between marketing, pricing, and sales to optimize strategies.
- **Predictive Analytics:** Using past data to predict the future.
- **Diagnostic Analytics:** Finding the root causes of problems such as declining sales or increased inventory.

d. Model Approach

Models are selected depending on specific objectives:

- **Regression models:** Predict revenue and order value using Random Forest or XGBoost.
- **Time series models:** Use ARIMA or Prophet to forecast demand over time.
- **Classification models:** Identify potential customers and predict churn.
- **Anomaly detection models:** Detect machine errors or data anomalies in the system.
- **Hybrid models:** Increase accuracy by combining multiple algorithms.

e. Output Design

The system's results will be presented in an easy-to-understand and user-friendly format:

- **BI Dashboard:** Uses Microsoft Power BI to display KPIs such as inventory, revenue, and production performance in real time.

- **Early warning system:** Sends notifications when inventory is low or when machinery shows signs of malfunction.
- **Strategic reporting:** Provides insights into product performance, market trends, and marketing campaign effectiveness.
- **Automated decision-making integration:** Several processes, such as reordering or production adjustments, can be automated based on data.

P7. Implement a data science solution to support decision making related to a real-world problem

1. Data Preparation

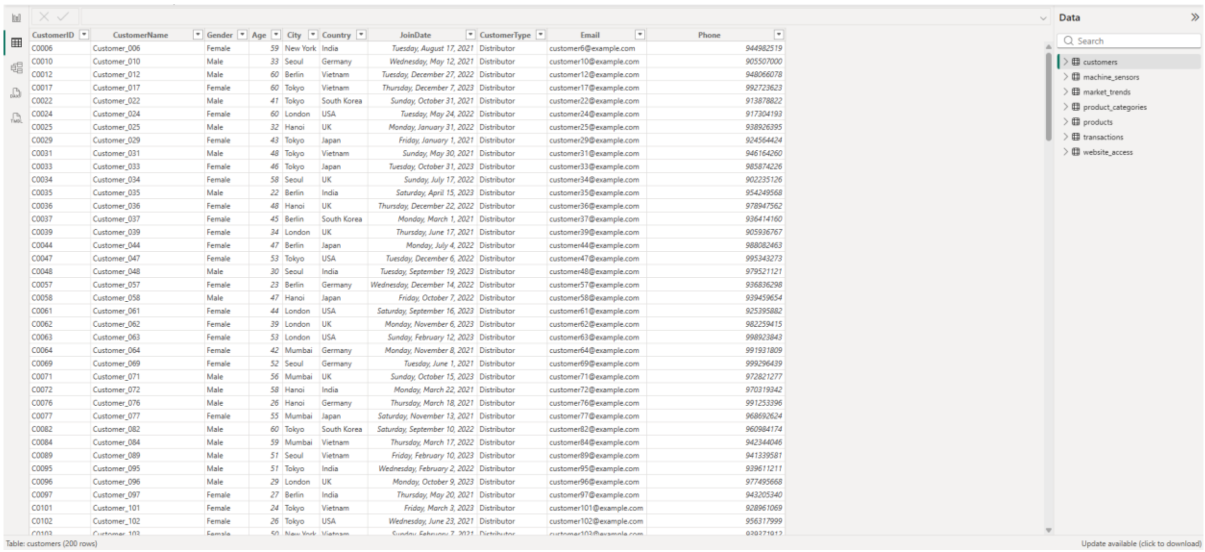
The dataset used in this project consists of multiple CSV files representing different aspects of Samsung Electronics operations, including customer data, transactions, product information, market trends, website interactions, and machine sensor data. These datasets are designed to simulate real-world business processes and support data-driven decision-making.

The key datasets include:

- **customers.csv:** Contains customer demographic information such as customer ID, location, and behavior-related attributes.
- **transactions.csv:** Records sales transactions, including product purchases, quantities, and transaction dates.
- **products.csv** and **product_categories.csv:** Provide details about products and their categories, supporting product-level analysis.
- **market_trends.csv:** Represents external factors such as seasonal demand and market fluctuations.
- **website_access.csv:** Tracks user interactions on digital platforms, including visits and engagement metrics.
- **machine_sensors.csv:** Simulates IoT data collected from manufacturing equipment, including temperature, vibration, power usage, and machine status, which is essential for predictive maintenance analysis.

These datasets are structured in CSV format and integrated to create a comprehensive view of business operations. The combination of transactional, behavioral, and sensor data enables the analysis of both customer demand and operational efficiency.

Before analysis, the data is prepared by ensuring consistency in format, handling missing values, and aligning key fields (such as IDs and timestamps) across datasets. This preparation step ensures that the data is reliable and suitable for further analysis and modeling.



CustomerID	CustomerName	Gender	Age	City	Country	JoinDate	CustomerType	Email	Phone
C0006	Customer_006	Female	59	New York	India	Tuesday, August 17, 2021	Distributor	customer6@example.com	944982519
C0010	Customer_010	Male	33	Seoul	Germany	Wednesday, May 12, 2021	Distributor	customer10@example.com	905507000
C0012	Customer_012	Male	60	Berlin	Vietnam	Tuesday, December 27, 2022	Distributor	customer12@example.com	948066078
C0017	Customer_017	Female	60	Tokyo	Vietnam	Thursday, December 7, 2023	Distributor	customer17@example.com	992723623
C0022	Customer_022	Male	41	Tokyo	South Korea	Sunday, October 31, 2021	Distributor	customer22@example.com	913878822
C0024	Customer_024	Female	60	London	USA	Tuesday, May 24, 2022	Distributor	customer24@example.com	917304193
C0025	Customer_025	Male	32	Hanoi	UK	Monday, January 31, 2022	Distributor	customer25@example.com	938926395
C0029	Customer_029	Female	43	Tokyo	Japan	Friday, January 1, 2021	Distributor	customer29@example.com	924564424
C0031	Customer_031	Male	48	Tokyo	Vietnam	Sunday, May 30, 2021	Distributor	customer31@example.com	946164260
C0033	Customer_033	Female	46	Tokyo	Japan	Tuesday, October 31, 2023	Distributor	customer33@example.com	985874226
C0034	Customer_034	Female	58	Seoul	UK	Sunday, July 17, 2022	Distributor	customer34@example.com	902235126
C0035	Customer_035	Male	22	Berlin	India	Saturday, April 15, 2023	Distributor	customer35@example.com	954249568
C0036	Customer_036	Female	48	Hanoi	UK	Thursday, December 22, 2022	Distributor	customer36@example.com	978947562
C0037	Customer_037	Female	45	Berlin	South Korea	Monday, March 1, 2021	Distributor	customer37@example.com	936414160
C0039	Customer_039	Female	34	London	UK	Thursday, June 17, 2021	Distributor	customer39@example.com	905936767
C0044	Customer_044	Female	47	Berlin	Japan	Monday, July 4, 2022	Distributor	customer44@example.com	988082463
C0047	Customer_047	Female	53	Tokyo	USA	Tuesday, December 6, 2022	Distributor	customer47@example.com	995343273
C0048	Customer_048	Male	30	Seoul	India	Tuesday, September 19, 2023	Distributor	customer48@example.com	979521121
C0057	Customer_057	Female	23	Berlin	Germany	Wednesday, December 14, 2022	Distributor	customer57@example.com	936836298
C0058	Customer_058	Male	47	Hanoi	Japan	Friday, October 7, 2022	Distributor	customer58@example.com	939459654
C0061	Customer_061	Female	44	London	USA	Saturday, September 16, 2023	Distributor	customer61@example.com	925395882
C0062	Customer_062	Female	39	London	UK	Monday, November 6, 2023	Distributor	customer62@example.com	982259415
C0063	Customer_063	Female	53	London	USA	Sunday, February 12, 2023	Distributor	customer63@example.com	998923843
C0064	Customer_064	Female	42	Mumbai	Germany	Monday, November 8, 2021	Distributor	customer64@example.com	991931809
C0069	Customer_069	Female	52	Seoul	Germany	Tuesday, June 1, 2021	Distributor	customer69@example.com	999296439
C0071	Customer_071	Male	56	Mumbai	UK	Sunday, October 15, 2023	Distributor	customer71@example.com	972821277
C0072	Customer_072	Male	58	Hanoi	India	Monday, March 22, 2021	Distributor	customer72@example.com	970319342
C0076	Customer_076	Male	26	Hanoi	Germany	Thursday, March 18, 2021	Distributor	customer76@example.com	991253396
C0077	Customer_077	Female	55	Mumbai	Japan	Saturday, November 13, 2021	Distributor	customer77@example.com	968692624
C0082	Customer_082	Male	60	Tokyo	South Korea	Saturday, September 10, 2022	Distributor	customer82@example.com	960984174
C0084	Customer_084	Male	59	Mumbai	Vietnam	Thursday, March 17, 2022	Distributor	customer84@example.com	942344046
C0089	Customer_089	Male	51	Seoul	Vietnam	Friday, February 10, 2023	Distributor	customer89@example.com	941339581
C0095	Customer_095	Male	51	Tokyo	India	Wednesday, February 2, 2022	Distributor	customer95@example.com	939611211
C0096	Customer_096	Male	29	London	UK	Monday, October 9, 2023	Distributor	customer96@example.com	977495668
C0097	Customer_097	Female	27	Berlin	India	Thursday, May 20, 2021	Distributor	customer97@example.com	943205340
C0101	Customer_101	Female	24	Tokyo	Vietnam	Friday, March 3, 2023	Distributor	customer101@example.com	928961069
C0102	Customer_102	Female	26	Tokyo	USA	Wednesday, June 23, 2021	Distributor	customer102@example.com	956317999

Figure 7. Data Preparation

2. Data Preprocessing

Data preprocessing is performed using Python to clean, transform, and integrate multiple datasets, ensuring data quality and consistency before analysis.

Data Cleaning:

Missing values in transactional data are handled using median imputation to maintain data integrity. For example, missing values in Quantity and UnitPrice are replaced with their respective median values. The TotalAmount field is then recalculated to ensure consistency:

```
df_trans['Quantity'] = df_trans['Quantity'].fillna(df_trans['Quantity'].median())
df_trans['UnitPrice'] = df_trans['UnitPrice'].fillna(df_trans['UnitPrice'].median())
df_trans['TotalAmount'] = df_trans['Quantity'] * df_trans['UnitPrice']
```

Figure 8. Data cleaning

Data Type Conversion:

Date fields are converted into a standard datetime format to support time-based analysis. This ensures consistency when performing forecasting and trend analysis:

```
df_trans['SaleDate'] = pd.to_datetime(df_trans['SaleDate'])
df_sensors['timestamp'] = pd.to_datetime(df_sensors['timestamp'])
```

Figure 9. Data type conversion

Data Integration (Merging Datasets):

Multiple datasets are combined to create a unified dataset for analysis. Transaction data is merged with product, category, and customer data to provide a comprehensive view of business operations:

```
master_df = df_trans.merge(df_prod[['ProductID', 'ProductName', 'ProductGroupID']], on='ProductID', how='left')
master_df = master_df.merge(df_cat[['ProductGroupID', 'GroupName']], on='ProductGroupID', how='left')
master_df = master_df.merge(df_cust[['CustomerID', 'CustomerName']], on='CustomerID', how='left')
```

Figure 10. Data integration

Data Standardization:

Key fields are aligned across datasets to ensure consistency, including product identifiers, customer IDs, and date formats. This step ensures that all datasets can be accurately linked and analysed together.

The result of the preprocessing process is a cleaned and integrated dataset, combining information such as transactions, customers, and products, ready for further analysis, modeling, and visualization.

	SaleID	CustomerID	ProductID	SaleDate	Quantity	UnitPrice	TotalAmount	DiscountPercent	PaymentMethod	Region	ProductNar	ProductGro	GroupName	CustomerName	PredictedRevenue
2	S0001	C0023	P006	1/5/2023	5	1076	5380	5	Bank Transfer	Europe	Samsung_P	G001	Smartphone	Customer_023	2913.2700485431888
3	S0002	C0075	P045	1/14/2024	5	290	1450	5	Cash	Global	Samsung_P	G001	Smartphone	Customer_075	3007.3150978593585
4	S0003	C0095	P009	1/31/2024	5	1011	5055	5	Cash	Europe	Samsung_P	G001	Smartphone	Customer_095	3012.364496480361
5	S0004	C0039	P034	1/13/2023	1	835	835	5	Cash	America	Samsung_P	G002	Tablet	Customer_039	2915.1635730260646
6	S0005	C0163	P018	1/1/2023	1	1043	1043	5	E-wallet	America	Samsung_P	G003	Television	Customer_163	2912.007698887938
7	S0006	C0027	P022	5/2/2023	2	783	1566	10	E-wallet	Global	Samsung_P	G010	Accessory	Customer_027	2945.459964752079
8	S0007	C0016	P010	5/24/2023	2	1079	2158	10	Credit Card	Asia	Samsung_P	G004	Laptop	Customer_016	2948.615838890206
9	S0008	C0060	P013	12/25/2023	5	1064	5320	5	Bank Transfer	Global	Samsung_P	G002	Tablet	Customer_060	2999.7409999278552
10	S0009	C0192	P012	2/18/2023	1	1111	1111	10	Cash	Global	Samsung_P	G009	Camera	Customer_192	2928.418244406196
11	S0010	C0020	P030	2/1/2023	5	1050	5250	10	E-wallet	Europe	Samsung_P	G009	Camera	Customer_020	2923.3688457851936
12	S0011	C0061	P020	6/11/2023	3	919	2757	5	E-wallet	America	Samsung_P	G007	Monitor	Customer_061	2953.665237511208
13	S0012	C0082	P009	4/30/2023	4	1011	4044	5	Credit Card	Europe	Samsung_P	G001	Smartphone	Customer_082	2944.1976150968285
14	S0013	C0035	P022	3/21/2023	5	783	3915	0	Credit Card	Europe	Samsung_P	G010	Accessory	Customer_035	2934.0988178548237
15	S0014	C0116	P028	7/28/2023	4	338	1352	5	Credit Card	America	Samsung_P	G010	Accessory	Customer_116	2966.288734063714
16	S0015	C0198	P037	6/7/2023	2	444	888	5	E-wallet	America	Samsung_P	G003	Television	Customer_198	2952.4028878559575
17	S0016	C0031	P004	2/18/2024	1	903	903	0	Bank Transfer	Asia	Samsung_P	G008	Audio	Customer_031	3014.8891957908622
18	S0017	C0056	P014	2/16/2023	3	382	1146	0	Cash	America	Samsung_P	G007	Monitor	Customer_056	2927.155894750945
19	S0018	C0102	P030	12/13/2023	5	1050	5250	0	Bank Transfer	America	Samsung_P	G009	Camera	Customer_102	2995.953950962103
20	S0019	C0045	P035	8/20/2023	1	376	376	5	Cash	Asia	Samsung_P	G004	Laptop	Customer_045	2971.969307512342
21	S0020	C0119	P006	3/4/2023	1	1076	1076	0	Credit Card	Asia	Samsung_P	G001	Smartphone	Customer_119	2930.9429437166973
22	S0021	C0200	P012	4/12/2023	2	1111	2222	0	E-wallet	America	Samsung_P	G009	Camera	Customer_200	2940.410566131077
23	S0022	C0176	P037	10/28/2023	1	444	444	10	Bank Transfer	Asia	Samsung_P	G003	Television	Customer_176	2985.223978892473
24	S0023	C0006	P032	11/27/2023	1	361	361	0	E-wallet	Europe	Samsung_P	G005	Wearable	Customer_006	2991.535727168726
25	S0024	C0024	P015	1/22/2023	3	480	1440	0	E-wallet	Asia	Samsung_P	G005	Wearable	Customer_024	2918.9506219918167
26	S0025	C0044	P050	2/25/2024	5	341	1705	5	E-wallet	Global	Samsung_P	G007	Monitor	Customer_044	3016.151545446113
27	S0026	C0008	P025	8/7/2023	4	1223	4892	10	Cash	America	Samsung_P	G004	Laptop	Customer_008	2969.4446082018408
28	S0027	C0055	P006	5/24/2023	2	1076	2152	5	E-wallet	Europe	Samsung_P	G001	Smartphone	Customer_055	2948.615838890206
29	S0028	C0009	P007	11/12/2023	5	525	2625	10	Cash	America	Samsung_P	G005	Wearable	Customer_009	2987.748678202974
30	S0029	C0128	P038	9/3/2023	4	535	2140	10	Credit Card	Asia	Samsung_P	G001	Smartphone	Customer_128	2975.7563564780935

Figure 11.

3. Data Analysis

Exploratory Data Analysis (EDA) is conducted to identify patterns, trends, and relationships within the dataset. This step helps transform raw data into meaningful insights that support business decision-making.

Sales Trend Analysis:

The transactional data is aggregated by date to analyze revenue trends over time. This allows the identification of growth patterns and seasonal fluctuations in sales performance.

```
# Aggregating revenue by date for the model
daily_rev = master_df.groupby('SaleDate')['TotalAmount'].sum().reset_index().sort_values('SaleDate')
daily_rev['day_index'] = np.arange(len(daily_rev))
```

Figure 12.

This analysis helps understand whether sales are increasing, stable, or declining, which is critical for forecasting and planning.

IoT Sensor Data Analysis (Machine Health):

Sensor data is analyzed to monitor machine conditions and detect potential issues.

```
machine_health = df_sensors.groupby('machine_id').agg({
    'sensor_temperature': 'mean',
    'sensor_vibration': 'max',
    'status': lambda x: (x == 'Warning').sum()
}).rename(columns={'status': 'WarningCount'}).reset_index()
```

Figure 13.

This analysis helps identify machines with abnormal temperature or frequent warnings, supporting predictive maintenance decisions.

Output:

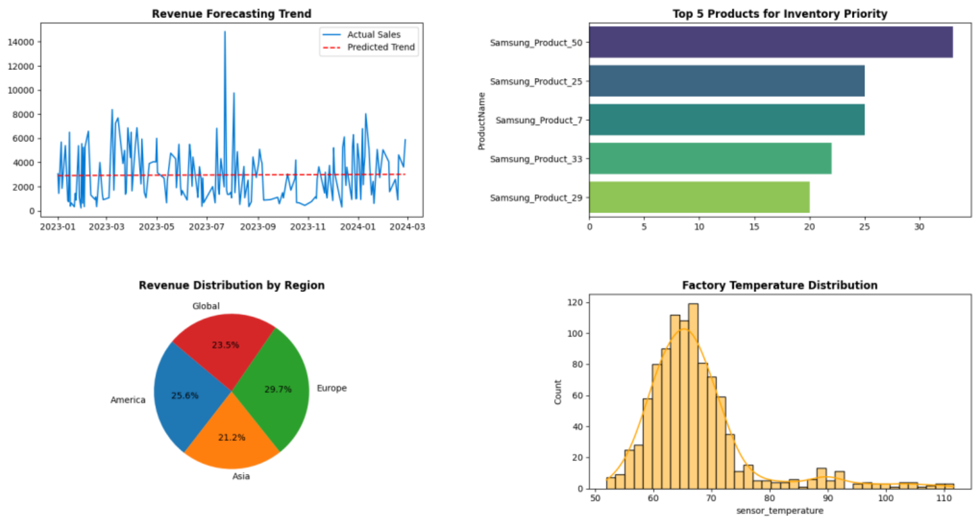


Figure 14. Output result

4. Data Visualization

Data visualization is conducted using graphical techniques to present key insights in a clear and user-friendly manner. These visualizations help decision-makers quickly understand patterns and trends within the data.

- **Revenue Forecasting Trend:** A line chart is used to compare actual revenue with predicted revenue over time. This allows the Operation Director to observe overall sales trends and evaluate future performance.
- **Top Products Analysis:** A bar chart is used to display the top-performing products based on total quantity sold. This helps identify high-demand products and supports inventory planning and stock prioritization.
- **Revenue Distribution by Region:** A pie chart is used to illustrate the proportion of total revenue contributed by each region. This enables the company to identify key markets and allocate resources more effectively.

- **Machine Temperature Distribution (IoT Analysis):** A histogram is used to analyze the distribution of machine temperature collected from IoT sensors. This helps detect abnormal patterns and supports maintenance planning.

5. Linking to Business Decisions

Based on the data analysis and visualizations, several data-driven recommendations can be proposed to support decision-making for the Operation Director.

Improve Demand Forecasting and Production Planning:

- **Recommendation:** Adjust production levels based on predicted revenue trends.
- **Justification:** The forecasting analysis shows the overall sales trend, allowing the company to anticipate future demand.
- **Suggested Action:** Increase production for periods with rising demand and reduce output during low-demand periods to avoid overstock.

Prioritize High-Demand Products:

- **Recommendation:** Focus on products with the highest sales volume.
- **Justification:** Product analysis identifies top-performing products that contribute significantly to total sales.
- **Suggested Action:** Allocate more inventory and resources to these products to minimize stockouts and maximize revenue.

Optimize Regional Distribution Strategy:

- **Recommendation:** Allocate resources based on regional sales performance.
- **Justification:** Revenue distribution analysis shows that some regions generate higher revenue than others.
- **Suggested Action:** Strengthen supply chains in high-performing regions and improve marketing or logistics strategies in lower-performing areas.

Enhance Predictive Maintenance (IoT):

- **Recommendation:** Monitor machines with abnormal temperature or frequent warnings.

- **Justification:** Sensor data analysis identifies machines at risk of failure, which can lead to production delays.
- **Suggested Action:** Schedule maintenance in advance to reduce downtime and avoid unexpected breakdowns.

M4. Make justified recommendations that support decision making related to a real-world problem

1. Overview of Analytical Results from P7

Based on data processing from various sources (Transactions, Products, IoT Sensors), we have gained a clear understanding of Samsung Electronics' operational situation:

- **Regarding Revenue Forecasting:** The Revenue Forecasting Trend chart shows significant volatility in actual sales. The current Linear Regression model is producing a fairly stable trend, but it hasn't yet caught up with the peaks of demand.
- **Regarding Product Performance:** We have identified the Top 5 strategic products, led by Samsung_Product_50 with the highest sales volume.
- **Regarding Market Distribution:** Europe is the largest revenue contributor (29.7%), closely followed by America (25.6%).
- **Regarding Machine Health (IoT):** Temperature chart analysis shows that most machines operate stably at 60–70°C, however, outliers have appeared exceeding 80°C, and even reaching 110°C.

2. Give some recommendation

Below are specific proposals to address the supply chain challenges I'm submitting to the Operations Director:

a. Optimizing Inventory for Key Product Lines

Recommendation: Prioritize allocating production resources and maintaining higher safety inventory levels for the Top 5 best-selling products, especially Samsung_Product_50. Simultaneously, implement real-time inventory tracking for this product group.

Justification: Data shows that demand for key products is higher and more stable than for the rest. Without a prioritization policy, rapid consumption can easily lead to shortages, especially during peak periods.

Expected Impact:

- Ensure stable revenue from strategic products.
- Minimize opportunity costs due to customer loss.
- Increase inventory turnover efficiency.

b. Implementing a Predictive Maintenance System Based on IoT Data

Recommendation: Immediately perform technical inspections on equipment with a MaintenanceAlert index of 1 and build a periodic maintenance schedule based on sensor data.

Justification: Temperature values exceeding safe limits (80°C – 110°C) indicate clear signs of equipment degradation. Without early intervention, the risk of complete system failure is very high.

Expected Impact:

- Minimize downtime.
- Save on repair and replacement costs.
- Increase machine lifespan and production stability.

c. Logistics Strategy Focused on Key Markets

Recommendation: Enhance real-time data tracking and optimize the transportation network in the European and US markets.

Justification: These two regions contribute over 55% of total revenue, so any disruption will significantly impact the entire business.

Expected Impact:

- Reduce lead time.
- Increase customer satisfaction.
- Strengthen competitive position in strategic markets.

D2. Evaluate the use of data science techniques against user and business requirements of an identified organisation

1. Organisation Overview

- **Organization Name:** Samsung Electronics
- **Size and Industry:** A multinational corporation operating in the manufacturing and distribution of consumer electronics, with a supply chain spanning multiple countries.
- **Project Background:** Samsung is facing increasing complexity in managing its global supply chain. Synchronizing production, distribution, and market demand is becoming more difficult as data volumes increase rapidly and market fluctuations continue.
- **The Role of Data:** Data plays a central role in connecting the various departments. From IoT sensor data in factories, sales data across regions, to customer behavior data – all contribute to a data-driven decision-making system.

2. Analysis of Business and User Requirements

Based on operational objectives, the requirements are clearly categorized:

Type of request	Request details
Business	Optimize inventory; Accurately forecast demand; Reduce operating costs; Ensure product quality and compliance with standards.
Internal users	Operations Director: needs a real-time dashboard; Production Head: needs early warning of machine failures; Data Analyst: needs clean and easily accessible data.
External users	Dealers/Distributors: require transparency regarding inventory and production plans; Customers: require products to be readily available.

Table 4. Analysis of Business and User Requirements

3. Data Science Techniques Applied

The system is built on a complete pipeline, integrating multiple techniques:

- **Data Processing and Cleaning:** Using Python (pandas) to process missing values using medians, remove duplicate data, and merge 5 data sources into a Master Dataset. Additionally, data is formatted to ensure consistency.

- **Descriptive Analytics:** Aggregating revenue by region, classifying products by performance, and identifying top-performing products.

- **Predictive Analytics:** Applying a Linear Regression model to determine revenue trends over time, providing a basis for production planning.

- **Machine Health Analysis (IoT):** Setting temperature warning thresholds (>80°C) and “Warning” status to detect potential equipment failures early.

- **Data visualization:** Using matplotlib and seaborn to build charts (Pie, Bar, Histogram), helping to transform data into easily understandable information.

4. Proof of Effectiveness and Alignment

The implemented techniques demonstrate a high degree of suitability to practical requirements:

Inventory Optimization: Analysis of the Top 5 products helps identify strategic items such as Samsung_Product_50, supporting production and distribution prioritization decisions.

Proactive Maintenance: The IoT alert system helps identify equipment at risk of failure early, shifting from a "repair only when broken" model to a "predictive" model.

Supply Chain Transparency: Analysis of revenue by region helps identify key markets (Europe and America), thereby optimizing logistics.

Accelerated Decision-Making: A visual dashboard allows the Operations Director to quickly grasp the situation without manual data processing.

5. KPIs and Quantitative Results

KPI	Quantitative results	Business impact
Revenue slope	+0.63 (Upward trend)	Verify the effectiveness of the current sales strategy.

Maintenance alert	15 devices need to be inspected.	Prevent sudden machine shutdowns in a timely manner.
Data processing	100% of the data has been cleaned.	Ensure that dashboard reports are always accurate and reliable.

Table 5. KPIs and Quantitative Results

6. Strengths and Limitations

Strengths:

- Fully automated process from cleaning to visualization, saving reporting time.
- Combines multiple data sources (IoT, Sales, Customers) for a multi-dimensional view.

Weaknesses:

- The Linear Regression model is still quite simple and doesn't keep up with extreme market fluctuations (sharp peaks in trend charts).
- Current IoT analysis is only based on fixed temperature thresholds and doesn't apply advanced anomaly detection algorithms.

7. Recommendations for Improvement

- **Applying advanced forecasting models:** Implementing Random Forest or Prophet algorithms to increase the accuracy of Samsung's seasonal forecasts.
- **Enhancing input data control:** Establishing data validation rules at the source system to reduce the error rate by 10% from the outset.
- **Implementing an automated alert system:** Connecting Python with the email/messaging API to immediately notify the technical department when machine temperatures exceed safe levels.

II. Conclusion

In conclusion, data and information have become fundamental drivers of modern business processes, enabling organizations to operate more efficiently, respond quickly to market changes, and make informed strategic decisions. Through effective data collection, processing, and analysis, businesses can gain valuable insights into customer behaviour, market trends, and internal performance, ultimately improving productivity and competitiveness.

This report has highlighted not only the practical value of data but also the broader implications associated with its use. Social, legal, and ethical considerations—such as data privacy, regulatory compliance, and responsible data handling—are essential factors that organizations must address. In addition, common threats including data breaches, malware, social engineering, and insider risks demonstrate the importance of implementing robust security measures at both individual and organizational levels.

From a strategic perspective, the application of data science and analytical techniques, particularly within organizations like Samsung Electronics, shows how data can significantly enhance demand forecasting, supply chain optimization, and decision-making processes. At the same time, the evaluation also emphasizes that ineffective data management can lead to serious consequences, including financial loss, operational disruption, and damage to brand reputation.

Therefore, to fully realize the benefits of data, organizations must adopt a comprehensive and well-structured data strategy. This includes investing in appropriate technologies, ensuring data quality, strengthening security frameworks, and maintaining transparency and ethical standards in data usage. By doing so, businesses can not only leverage data as a competitive advantage but also build long-term trust with customers and stakeholders, ensuring sustainable growth in an increasingly data-centric world.

IV. Evaluation

Through this module, I have developed a clear understanding of the importance of data and information in modern business environments. I now recognize how data supports decision-making, improves operational efficiency, and helps organizations better understand customers and market trends. In particular, analysing real-world scenarios such as Samsung Electronics has helped me see how data is applied practically in areas like supply chain management, demand forecasting, and performance monitoring.

One of the key things I have learned is that data is not only valuable but also comes with responsibilities. I now understand both the positive and negative implications of using data, including social, legal, and ethical issues such as privacy, security, and transparency. This has helped me develop a more balanced perspective, where data should be used effectively but also responsibly.

I also gained knowledge about how data is generated, processed, and transformed into meaningful insights using different tools and techniques. Learning about data science has been particularly valuable, as it shows how advanced methods can be used to solve complex business problems and support strategic decision-making.
However, I found the data science aspect of the module quite challenging. Since I had no prior experience with data science concepts or tools, understanding topics such as data processing pipelines, machine learning models, and analytical techniques required extra effort. At times, it was difficult to fully grasp how these models work in practice, especially when dealing with technical details.

Despite these challenges, this module has been a valuable learning experience. It has provided me with foundational knowledge that I can continue to build on in the future. Moving forward, I plan to improve my skills in data analysis and data science by practicing more with tools like Python and Power BI, so that I can better apply these concepts in real-world business situations.

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Tools used to assist in the report writing process: Chat GPT + Gemini

VI. Appendix

Power BI Dashboard: https://github.com/Lyra-Everlyn/Samsung-Electronics-Dashboard-Power-BI.git